Atmospheric Aerodynamics

Laboratory Session 2

Air-Water Flow Modelling: Hydraulic Jump over an Ogee Weir

Master's Degree in Computational Fluid Dynamics

Department of Hydraulic and Environmental Engineering

Universitat Politècnica de València

Author: Nieves Castellanos Arnáez

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Introduction

The hydraulic behaviour of free-surface flows is governed by the interplay between inertial and gravitational forces. Understanding the transitions between subcritical and supercritical regimes is central to designing hydraulic structures like weirs and stilling basins. This report documents the implementation and analysis of a two dimensional Computational Fluid Dynamics (CFD) model of flow over an ogee weir, including a downstream hydraulic jump.

The simulation was performed in OpenFOAM using the `interFoam` solver for two-phase (air-water) flow and the RANS k-ω SST turbulence model. This case is valuable as it encompasses a subcritical approach flow, a critical transition at the crest, a supercritical chute, and massive energy dissipation via a hydraulic jump stabilised by an end sill. The subsequent sections describe the case setup, boundary conditions, and provide a critical analysis of the simulated quasi-steady flow features.

Case Description

Physical scenario

The scenario consists of subcritical flow approaching an ogee weir from the left, passing over it, and forming a supercritical jet on the downstream face of the structure. Further downstream, the flow decelerates abruptly, giving rise to a hydraulic jump before recovering a subcritical regime at the outlet.

The key flow parameters are summarised in table 1. The weir height is $H = 0.117\,\text{m}$ and its width is $a = 3H = 0.351\,\text{m}$. Downstream of the chute, an end sill consisting of a row of teeth ($0.050\,\text{m}$ wide, $0.025\,\text{m}$ tall) begins at $x = 0.90\,\text{m}$. This feature is intended to promote energy dissipation and stabilise the hydraulic jump.

Table 1: Hydraulic parameters at inlet and outlet.

Location	Depth h (m)	Velocity U (m/s)	Froude number Fr	Regime
Inlet	0.200	0.25	0.13	Subcritical
Outlet	0.154	0.325	0.26	Subcritical

It is worth noting that both the inlet and outlet are subcritical ($Fr < 1$), while the flow accelerates to supercritical conditions ($Fr > 1$) over the weir crest. The energy dissipated in the hydraulic jump effectively bridges this transition back to the subcritical regime imposed at the outlet.

Geometry

The computational domain is rectangular and spans $x \in [-2a,\, 5a]$ and $z \in [0,\, 2.5H]$. The ogee weir crest is located at the origin ($x = 0$, $z = 0$), with the channel floor at $z = 0$.

The weir profile is defined parametrically as:

$$x(\xi) = \frac{1}{2}\xi\left(1 + \frac{a^2}{\xi^2 + m^2(a^2 - \xi^2)}\right), \tag{1}$$

$$z(\xi) = \frac{1}{2}m\sqrt{a^2 - \xi^2}\left(1 - \frac{a^2}{\xi^2 + m^2(a^2 - \xi^2)}\right), \tag{2}$$

where $\xi \leq |a|$ and the shape parameter m is:

$$m = \frac{H}{a} + \sqrt{\left(\frac{H}{a}\right)^2 + 1}. \tag{3}$$

Numerical Implementation

Mesh

The mesh is a structured, body fitted grid of quadrilateral cells with a uniform size of $\Delta x = 0.005\,\mathrm{m}$ throughout the domain. This cell size was selected as a baseline compromise between resolution and computational cost. The body fitted approach ensures that the ogee weir profile and the end sill teeth are adequately represented without the need for cut cell or immersed boundary techniques.

Boundary conditions

The boundary conditions applied at each patch are summarised in table 2. At the inlet, a hydrostatic pressure profile consistent with $h_1 = 0.20\,\mathrm{m}$ is imposed together with a uniform horizontal velocity of $U_1 = 0.25\,\mathrm{m/s}$. At the outlet, the hydrostatic pressure corresponding to $h_2 = 0.154\,\mathrm{m}$ is prescribed and a zero gradient condition is applied to velocity.

The top boundary (atmosphere) uses a Dirichlet condition for pressure ($p = 0$) and a zero gradient condition for velocity. This setup naturally allows air to enter or leave the domain as required by the flow dynamics, without imposing artificial constraints on the free surface. The bottom and weir surfaces are treated as no-slip walls with a wall function for the turbulence variables, which avoids the need for excessive near wall refinement while still capturing the shear driven effects. Front and back patches are set to `empty` in accordance with the pseudo 2D nature of the model.

Table 2: Boundary condition summary (S.I. units).

Patch	Type	p	U	k, ω	F
Left	Inlet	$\nabla p = 0$	$U_x = 0.25$	$\xi = 0.001$	$F = 1$
Right	Outlet	$p = \gamma h_2$	$\nabla U = 0$	$\nabla \xi = 0$	$\nabla F = 0$
Bottom	No-slip	$\nabla p = 0$	$U = 0$	Wall function	$\nabla F = 0$
Top	Atmosphere	$p = 0$	$\nabla U = 0$	$\nabla \xi = 0$	$\nabla F = 0$

Initial conditions

The domain is initialised with the region upstream of the weir and below $z = h_1$ filled with water ($F = 1$), and the remainder set to air ($F = 0$), consistently with the expected steady-state configuration. All velocity components, pressure, k, and ω are initialised to zero or their minimum prescribed values ($k_0 = 0.001$, $\omega_0 = 0.001$).

Fluid properties

Both phases are treated as Newtonian and incompressible. The relevant properties are listed in table 3. Surface tension between air and water is $\sigma = 0.07\,\mathrm{N/m}$ and gravitational acceleration is $g = 9.81\,\mathrm{m/s^2}$.

Table 3: Fluid properties of each phase.

Property	Air	Water
ρ (kg/m^3)	1.25	1000
ν (m^2/s)	1.5×10^{-5}	1.0×10^{-6}

Turbulence modelling

Turbulence is modelled with the RANS k-ω SST closure. This model is particularly suited to the present case because it combines the robustness of the k-ω formulation near walls with the free stream insensitivity of the k-ε model in the outer flow, making it reliable across the range of flow conditions encountered here (from the wall-bounded approach flow to the highly turbulent hydraulic jump roller).

Solver and numerical schemes

The case is solved with the PIMPLE algorithm in transient mode. An adjustable time step is used, constrained by a Courant Friedrichs Lewy (CFL) condition of $Co \leq 0.3$. Time integration uses an implicit second order scheme. All spatial discretisation schemes are second-order accurate. The simulation is run to $t_s = 15\,\mathrm{s}$.

Results and Discussion

Flow depth evolution

The fig. 1 shows the temporal evolution of flow depth at the four monitoring cross-sections: $x = -0.5\,\text{m}$, $x = 0.0\,\text{m}$, $x = 0.5\,\text{m}$, and $x = 1.0\,\text{m}$.

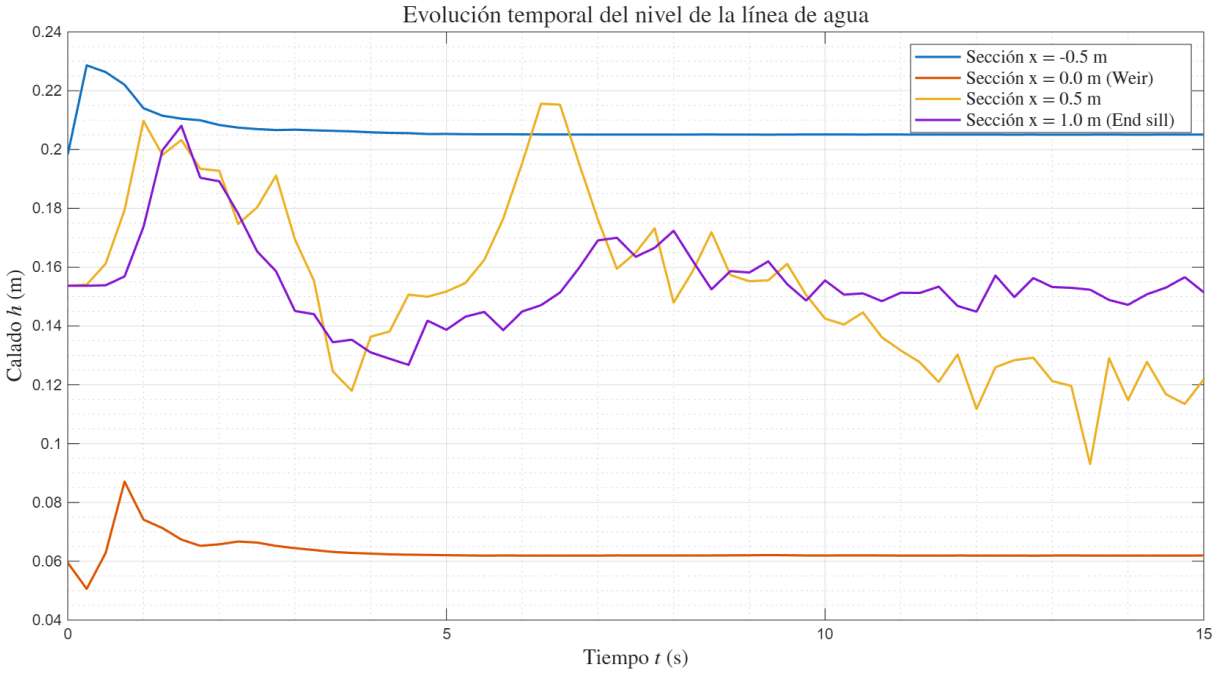


Figure 1: Temporal evolution of water depth at $x = -0.5$, 0.0, +0.5, and +1.0 m.

The section at $x = -0.5\,\text{m}$ (blue) exhibits a brief initial transient peak of approximately 0.23 m before converging to a stable quasi-steady value close to 0.205 m, consistent with the imposed inlet depth of 0.20 m. The rapid stabilisation of this upstream section confirms that the subcritical inlet boundary condition is well imposed and that backwater effects from the weir do not significantly disturb the approach flow after the initial transient.

At the weir crest ($x = 0.0\,\text{m}$, orange), the depth quickly settles to approximately 0.06 m after an early peak near 0.09 m. This low value is consistent with critical flow conditions at the crest, where the flow depth must be close to the critical depth $h_c = (q^2/g)^{1/3}$. The fast convergence of this section confirms that the weir acts as a hydraulic control from early in the simulation.

The two downstream sections, $x = 0.5\,\text{m}$ (yellow) and $x = 1.0\,\text{m}$ (purple), display much larger and more persistent fluctuations throughout the simulation, particularly between $t = 5\,\text{s}$ and $t = 10\,\text{s}$. This unsteadiness is physically meaningful: both sections lie within or immediately downstream of the hydraulic jump roller, a region inherently characterised by unsteady turbulent mixing. By $t = 15\,\text{s}$, the depth at $x = 1.0\,\text{m}$ oscillates around approximately 0.15 m, which is in good agreement with the prescribed outlet depth of

0.154 m. The section at $x = 0.5$ m still shows noticeable fluctuations at the end of the simulation, suggesting that the jump has not yet reached a fully periodic quasi-steady state within the 15 seconds simulated. Longer simulation times would be required to obtain truly time-averaged statistics in this region.

Flow structure and qualitative observations

The figs. 2 and 3 present the velocity magnitude field and the velocity streamlines at quasi-steady state ($t = 15$ s), while fig. 4 shows the fluid fraction field.

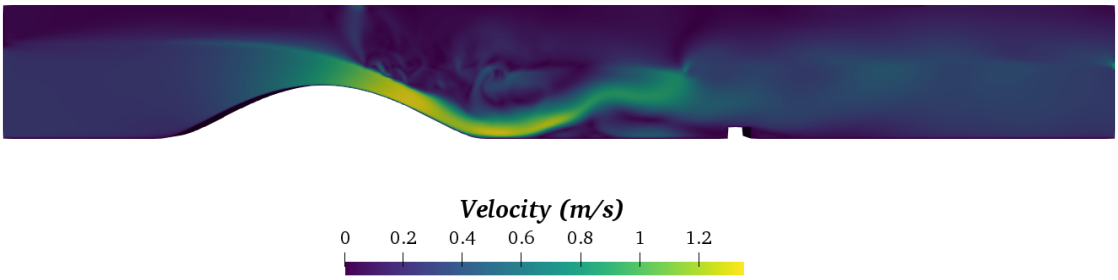


Figure 2: Velocity magnitude field at $t = 15$ s.

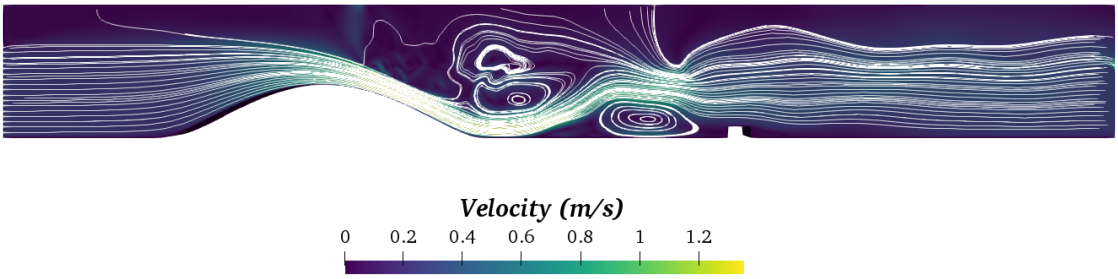


Figure 3: Velocity magnitude (colour) and streamlines at $t = 15$ s.

Several physically meaningful phenomena are clearly identified in the results.

Gradually varied flow (GVF). Upstream of the weir, the flow is subcritical and the velocity is low and uniform (deep blue in fig. 2). The water surface drops smoothly as the flow accelerates towards the weir crests. The streamlines in this region are nearly horizontal and parallel, confirming quasi dimensional behaviour.

Critical transition and supercritical chute. At the weir crest the flow passes through critical conditions and subsequently accelerates sharply down the downstream face. The velocity magnitude reaches values above 1.2 m/s in this region (yellow band in fig. 2), representing a roughly fivefold increase over the inlet velocity. This is a classic rapidly varied flow (RVF) region.

Hydraulic jump and roller. Immediately downstream of the chute, the high velocity jet impinges on the slower downstream water, forming a hydraulic jump. The streamlines in fig. 3 reveal two clearly defined recirculation vortices: a large surface roller occupying most of the jump region, and a secondary counter rotating vortex near the channel floor just upstream of the end sill. These structures are the main agents of energy dissipation in the system. The end sill further disrupts the near bed flow, enhancing dissipation and helping to stabilise the jump position.

Deaeration and recovery. Downstream of the jump, the streamlines progressively realign with the horizontal as the flow decelerates back to a subcritical regime, consistent with the outlet boundary condition.

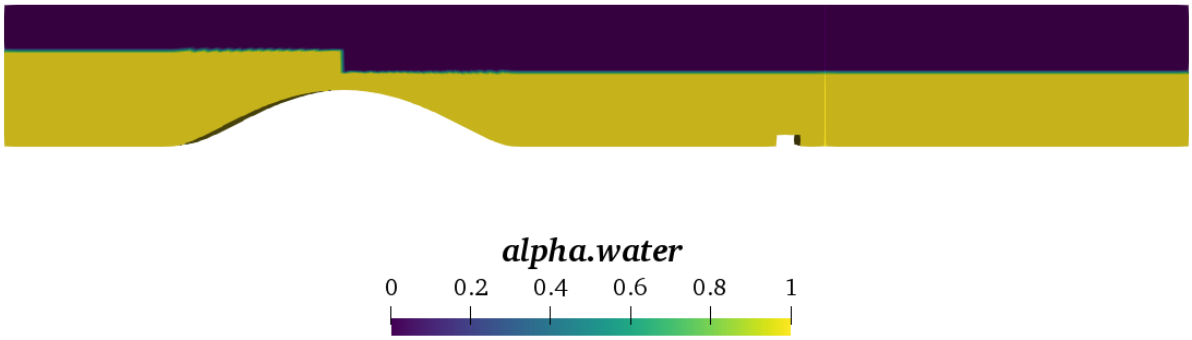


Figure 4: Fluid fraction field ($F = 1$ water, $F = 0$ air) at quasi-steady state. Yellow regions correspond to water; dark regions to air.

The fluid fraction field in fig. 4 shows a well-defined free surface upstream of the weir and a notably thinner water layer over the crest and along the chute. The interface between air ($F \approx 0$, dark) and water ($F \approx 1$, yellow) is sharp upstream and across the weir profile, which confirms adequate mesh resolution for the VOF interface in those regions. At the end sill location a small air pocket is visible on the lee side of the tooth, consistent with flow separation at this sharp geometric feature. Downstream of the jump the water layer deepens again towards the outlet depth.

It is noteworthy that the present snapshot does not show pronounced air entrainment within the roller, likely because the mesh resolution of $\Delta x = 0.005\,\text{m}$ is insufficient to resolve the fine scale interfacial structures responsible for bubble entrainment. A finer mesh in the jump region would be expected to capture aeration more accurately.

Pressure field

fig. 5 shows the pressure distribution at quasi-steady state.

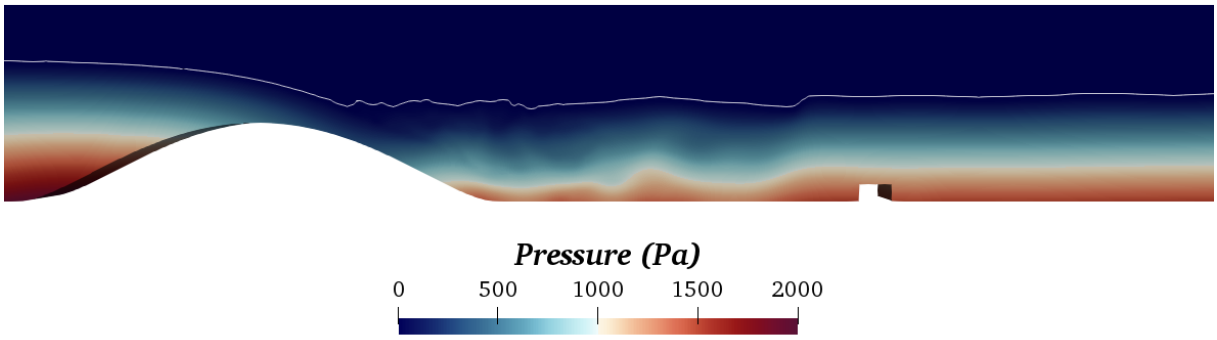


Figure 5: Gauge pressure field (Pa) at quasi-steady state. The white contour indicates the approximate free-surface location ($F = 0.5$).

The pressure field is consistent with hydrostatic conditions throughout most of the domain. Upstream of the weir, pressure increases linearly with depth as expected, reaching values around 2000 Pa at the channel floor (corresponding to $\rho g h_1 \approx 1962$ Pa). Over the weir crest the pressure drops to near zero, reflecting the low water depth at this location. Along the downstream chute and in the jump region, pressure gradients are clearly non hydrostatic due to the strong vertical accelerations and curvature of the streamlines. The near bed pressure in the hydraulic jump zone shows intermediate values that recover gradually towards hydrostatic conditions further downstream. The white free surface contour confirms that the pressure field is continuous and physically consistent across the air–water interface.

Vorticity field

In fig. 6 it is possible see the vorticity magnitude field ($|\omega_y|$, in s^{-1}) at quasi-steady state.

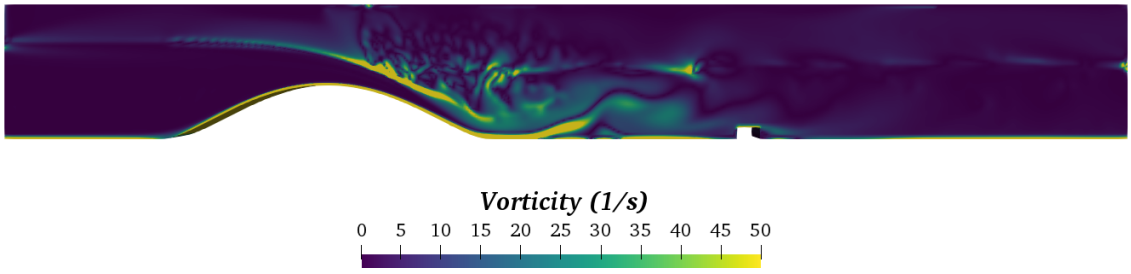


Figure 6: Vorticity magnitude field (s^{-1}) at quasi-steady state. The colour scale ranges from 0 to 744 s^{-1}.

The vorticity field confirms that rotational motion is confined to the region of the weir and the hydraulic jump. Upstream and far downstream, the flow is nearly irrotational as expected for a slowly moving subcritical current. High vorticity concentrations are visible

along the shear layer that develops on the downstream face of the weir and at the toe of the hydraulic jump, where the supercritical jet meets the slower roller above it. The end sill also generates a localised zone of elevated vorticity due to the abrupt flow separation at its upstream face.

Hydraulic jump roller length

The roller length L_r is estimated from the streamline field in fig. 3 as the horizontal distance between the toe of the jump (where the supercritical jet first separates from the channel floor) and the downstream reattachment point of the surface recirculation. Based on visual inspection of the results, the roller extends approximately from $x \approx 0.35\,\mathrm{m}$ to $x \approx 0.85\,\mathrm{m}$, yielding:

$$L_r \approx 0.50\,\mathrm{m} \tag{4}$$

This value can be compared with the empirical correlation of Hager [2], who proposes $L_r/h_1 \approx 6(Fr_1 - 1)$ for the roller length in a classical hydraulic jump. The exact pre jump Froude number Fr_1 at this location would need to be extracted from the simulation data for a rigorous comparison; however, the order of magnitude is consistent with the observed roller extent.

Wall y^+ analysis

fig. 7 shows the distribution of the dimensionless wall coordinate y^+ along the channel floor and weir surface at quasi-steady state.

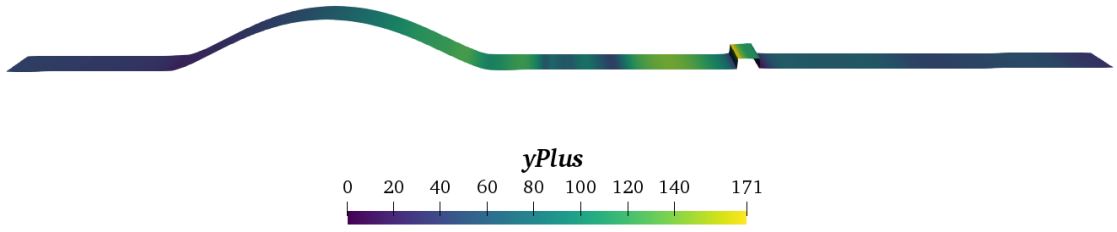


Figure 7: y^+ distribution along the channel floor and weir surface.

The y^+ values range from approximately 0 in the low velocity upstream region and in the recirculation zones downstream of the jump, up to a maximum of about 171 in the high velocity region along the weir chute. The majority of the wall surface (particularly the flat channel floor and the weir profile) shows y^+ values in the range $30 \lesssim y^+ \lesssim 171$, which falls within the validity range of the log-law region ($30 \lesssim y^+ \lesssim 300$) where wall functions are applicable. This confirms that the chosen mesh size of $\Delta x = 0.005\,\mathrm{m}$ is appropriate for the wall function treatment employed.

However, several regions exhibit $y^+ < 30$: the far upstream flat bed section (very low near wall velocity), and the zones immediately downstream of the hydraulic jump roller where the flow re accelerates slowly. In these regions, the wall function is applied outside its strict validity range. For the present engineering level study, this is acceptable, but a mesh refinement near the wall or the use of a low Reynolds number turbulence model in these areas would improve the near wall accuracy. The peak values near the end sill (visible as the localised maximum in the figure) reflect the high velocity impingement of the supercritical jet at this geometric feature, where the wall shear stress is at its highest.

Conclusions

This CFD study successfully modelled the complex free surface flow over an ogee weir and the subsequent hydraulic jump using OpenFOAM's `interFoam` solver. The temporal evolution of the water depth confirmed that the upstream and weir crest sections reached a quasi-steady state rapidly, while the downstream jump region exhibited expected turbulent unsteadiness, eventually converging towards the prescribed outlet depth of 0.154 m. The model accurately captured the transition from subcritical to supercritical flow, with the velocity and streamline fields clearly resolving the strong energy dissipation mechanisms: the primary surface roller (with an estimated length of $L_r \approx 0.50\,\text{m}$) and the secondary recirculation induced by the end sill. The pressure field correctly reflected hydrostatic conditions in subcritical zones and non hydrostatic variations in rapidly varied regions. Finally, the y^+ analysis validated the use of wall functions for the k-ω SST turbulence model across most of the domain ($30 \lesssim y^+ \lesssim 171$), although minor deviations in low velocity regions highlight areas where future local mesh refinement could further improve near wall accuracy.

References

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